

First Terminal Examination - 2079

Class : XI

Subject: Computer Science (4271)

Full Marks: 50

Time: 2 hrs.

Set - A

Pass Marks: 20

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

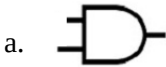
Group 'A'

Multiple choice Questions

[9 × 1 = 9]

Rewrite the best alternative.

- The smallest unit of data in computer is called
 - Bytes
 - Bits
 - Nibble
 - Kilobyte
- The speed of computer can be measured in term of
 - Megabits and Gigabits
 - Megabytes and Gigabytes
 - Megahertz and Gigahertz
 - All of the above
- How to represent Boolean $F(x) = (x)'$ in logic gate?



- One pico second means
 - 10^{-3} of seconds
 - 10^{-6} of seconds
 - 10^{-9} of seconds
 - 10^{-12} of seconds
- In computer system, 1 nibble equals to
 - 1 bits
 - 8 bytes
 - 4 bits
 - 8 bits

6. The binary equivalent of $(0.12)_8$ is
- | | |
|-------------|-------------|
| a. 1010 | b. 0.010110 |
| c. 0.001010 | d. 0.001100 |
7. 2's complement of 1011 is
- | | |
|---------|---------|
| a. 0100 | b. 0101 |
| c. 0111 | d. 1100 |
8. The NOR gate output will be high if the two inputs are
- | | |
|--------|--------|
| a. 0 0 | b. 0 1 |
| c. 1 0 | d. 1 1 |
9. $(E7)_{16} = (?)_8$
- | | |
|--------|--------|
| a. 743 | b. 167 |
| c. 176 | d. 347 |

Group 'B'

Give short answer to the following questions.

[5 × 5 = 25]

1. Differentiate between third and fourth generation computer
2. Subtract $(110101)_2$ from $(1101)_2$ using 1's and 2's complement method.
3. How to use computer in education and science and engineering fields. Explain.
4. What is logic gate? Explain about NAND and NOR gate.
5. Convert following numbers as indicated in the bracket.
 - a) $(573.12)_8 = (?)_{16}$
 - b) $(123.625)_{10} = (?)_{16}$

Group 'C'

Give long answer to the following question.

[2 × 8 = 16]

6. Explain any eight characteristics of computer.
7. Prove the associative law, distributive law and commutative law with their truth table and also draw graphical symbol.

*****Best of Luck*****

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Group 'A'

Multiple choice Questions

[9 × 1 = 9]

Rewrite the best alternative.

- A computer is a/an device that processes data and converts it into information.
 - Electrical
 - Mechanical
 - Electronic
 - All of the above
- Bio-chips are related to which generation of computer?
 - Second Generation
 - Third Generation
 - Fourth Generation
 - Fifth Generation
- One nano second means
 - 10^{-3} of seconds
 - 10^{-6} of seconds
 - 10^{-9} of seconds
 - 10^{-12} of seconds
- Which one of the following is universal gate?
 - OR, AND NOR
 - NOR, NAND
 - OR, NOR
 - AND, NAND
- Which of the following unit is responsible for converting the data received from the user into a computer understandable format?
 - Processing Unit
 - Input Unit
 - Output Unit
 - Memory Unit
- The binary equivalent of $(0.73)_{16}$ is
 - 01110011
 - 0.11101100
 - 0.0110011
 - 0.01110011

7. 2's complement of 10111 is
 - a. 01000
 - b. 01100
 - c. 01001
 - d. 01001
8. The NAND gate output will be low if the two inputs are
 - a. 0 0
 - b. 0 1
 - c. 1 0
 - d. 1 1
9. $(734)_8 = (?)_{16}$
 - a. 1EC
 - b. C1D
 - c. 1CD
 - d. 1DC

Group 'B'

Give short answer to the following questions.

[5 × 5 = 25]

1. Subtract $(1010)_2$ from $(111100)_2$ using 1's and 2's complement method.
2. Describe at least five ways in which information technology can help students studying subjects other than computing.
3. State and prove De-morgan's law with the help of truth table and also draw the logical diagram.
4. Why computer is known as versatile and diligent machine? Explain.
5. Convert following numbers as indicated in the bracket.
 - a) $(333.BC)_{16} = (?)_8$
 - b) $(63.52)_8 = (?)_2$

Group 'C'

Give long answer to the following question.

[2 × 8 = 16]

6. Explain the evolution of computer describing the technologies used in different generations.
7. Explain X-OR and X-NOR gate with their graphical symbol, truth table, Boolean expression, venn-diagram, logical structure and definition.

****Best of Luck****